



DebugFS

DebugFS is a simple memory-based filesystem, designed specifically to debug Linux kernel code, and not to be confused with the DebugFS filesystem utility. Introduced by Greg Kroah-Hartman in December 2004, DebugFS helps kernel developers export large amounts of debug data into user-space. This article gives a good introduction to DebugFS and its application programming interface, along with sample code. The intended audience are kernel and device driver developers with some knowledge of Linux kernel development. The experienced can use it to refresh their skills as well.

Greg Kroah-Hartman introduced DebugFS to the Linux kernel through his post to the Linux kernel mailing list (LKML) (<http://marc.info/?l=linux-kernel&m=110264016924974&w=2>) in December 2004. DebugFS helps kernel developers export large amounts of debug data into user-space. It has no rules about any information that can be put in, unlike */proc* and */sysfs*, and hence is encouraged for use in

debugging. Using *printfk* can suffice when reading values—but it does not allow developers to change values of variables from user-space.

Prerequisites

A test system with kernel development environment requires Linux kernel source code for reference. Pretty much any distribution with a kernel version higher than 2.6.10-rc3 supports DebugFS, but the newer the kernel, the more API support you get.